

CENTRAL BOARD OF SECONDARY EDUCATION

HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)

SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 7: WORK, ENERGY, AND SIMPLE MACHINES [SET C - NCERT EXEMPLAR SPECIAL]

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25

PAPER CODE: 086/CH07/SET-C

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET C - NCERT EXEMPLAR & CRITICAL THINKING):

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15)**: NCERT Exemplar based Multiple Choice Questions with single correct option.
3. **Section B (Q16 – Q20)**: Deep Assertion-Reasoning based questions testing scientific cause-and-effect.
4. **Section C (Q21 – Q25)**: Case-Study / Laboratory Experimental Investigation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: NCERT EXEMPLAR MCQS (Q.1 TO Q.15)

[15 MARKS]

Q.1 A roller coaster car of mass m enters a vertical circular loop of radius R . What is the minimum velocity required at the lowest point of the track to complete the loop without falling off at the apex (g is acceleration due to gravity)?

- (A) $\sqrt{gR}$ (B) $\sqrt{2gR}$
(C) $\sqrt{5gR}$ (D) $\sqrt{3gR}$

Q.2 A force $\vec{F} = (3\hat{i} + 4\hat{j})$ N acts on a particle, displacing it by $\vec{s} = (5\hat{i} + 2\hat{j})$ m. What is the total work done?

- (A) 15 J (B) 23 J
(C) 8 J (D) 35 J

Q.3 A 1000 kg car travelling at 72 km/h (20 m/s) has its kinetic energy converted into heat by friction brakes to stop. How much thermal energy is dissipated in the brakes?

- (A) 100 kJ (B) 200 kJ
(C) 400 kJ (D) 50 kJ

Q.4 A spring with spring constant $k = 500$ N/m is compressed by 4 cm (0.04 m). What is the magnitude of the restoring force exerted by the spring?

- (A) 20 N (B) 200 N
(C) 2 N (D) 0.4 N

Q.5 A pump is rated at 2.0 kW. How many liters of water can it lift in 10 minutes to an overhead tank 10 m high ($g = 10 \text{ m/s}^2$, density of water $= 1 \text{ kg/L}$)?

- (A) 6,000 L (B) 12,000 L
(C) 1,200 L (D) 24,000 L

Q.6 Which of the following is a non-conservative force where work done depends on the actual path taken?

- (A) Gravitational force (B) Electrostatic force
(C) Kinetic friction force (D) Ideal spring restoring force

Q.7 A 2 kg metal ball is released from rest from a height of 5 m above a sandpit. It penetrates 0.10 m into the sand before stopping. What is the average retarding force exerted by the sand ($g = 10 \text{ m/s}^2$)?

- (A) 1000 N (B) 1020 N
(C) 500 N (D) 200 N

Q.8 A wheel and axle machine has a wheel radius of 40 cm and an axle radius of 8 cm. What is the ideal mechanical advantage (velocity ratio)?

- (A) 5 (B) 0.2
(C) 4 (D) 32

Q.9 An object of mass m has a momentum p . Its kinetic energy is given by:

- (A) pm (B) $\frac{p^2}{2m}$
(C) $\frac{2p^2}{m}$ (D) $\frac{p}{2m}$

Q.10 A constant force F accelerates an object from rest to speed v over time t . The instantaneous power delivered by the force at time t is:

- (A) Fv (B) $\frac{1}{2} Fv$
(C) $2 Fv$ (D) $\frac{Fv}{t}$

Q.11 What is the average power delivered in the above problem?

- (A) Fv (B) $\frac{1}{2} Fv$
(C) $2 Fv$ (D) Fv^2

Q.12 A hydroelectric turbine operates with 85% efficiency under a head of 100 m. If water flows through it at 10,000 kg/s ($g = 10 \text{ m/s}^2$), what electrical power is generated?

- (A) 8.5 MW (B) 10 MW
(C) 1.5 MW (D) 85 MW

Q.13 A simple machine lifts a load of 400 N through 2 m when an effort of 100 N moves through 10 m. What is the efficiency of the machine?

- (A) 80% (B) 75%
(C) 90% (D) 85%

Q.14 A body of mass 1 kg is thrown vertically upward with 100 J of kinetic energy. At what height above the launch point will its kinetic energy equal its potential energy ($g = 10 \text{ m/s}^2$)?

- (A) 2.5 m (B) 5.0 m
(C) 10.0 m (D) 1.25 m

Q.15 Which class of lever has the mechanical advantage always less than 1 ($MA < 1$)?

- (A) Class 1 lever (B) Class 2 lever
(C) Class 3 lever (D) Inclined plane

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): Work done by gravity on an object sliding down a frictionless incline of height h equals the work done dropping it vertically through height h .

Reason (R): Gravitational force is a conservative force, and work done between two vertical levels depends only on initial and final heights, not the path taken.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.17 Assertion (A): A heavier truck and a light car having identical kinetic energies have different momenta.

Reason (R): Momentum is proportional to $\sqrt{m}$ for a given constant kinetic energy ($p = \sqrt{2m E_k}$).

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.18 Assertion (A): A machine cannot multiply both force and speed simultaneously.

Reason (R): Work output cannot exceed work input ($\text{Load} \times d_L \leq \text{Effort} \times d_E$); if force is multiplied ($MA > 1$), distance and speed must be compromised ($VR < 1$).

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.19 Assertion (A): When a gas expands isothermally against atmospheric pressure, it does positive work.

Reason (R): The force exerted by the expanding gas acts in the same direction as the displacement of the piston.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.20 Assertion (A): A stationary satellite in orbit consumes rocket fuel continuously to do gravitational work.

Reason (R): The gravitational pull of the Earth continuously accelerates the satellite inward.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL INVESTIGATION & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL INVESTIGATION:

Case Study: Roller Coaster Gravitational & Kinetic Energy Dynamics

A roller coaster car of total mass 500 kg starts from rest at the top of an initial lift hill at Height A ($h_A = 45$ m) above the ground. The car coasts down into a valley at ground level ($h = 0$) and then ascends into a second smaller hill of Height B ($h_B = 25$ m). Neglect friction and air resistance ($g = 10 \text{ m/s}^2$).

Q.21 What is the gravitational potential energy of the roller coaster car at the peak of the initial hill (Height A)?

- (A) 22,500 J (B) 225,000 J (225 kJ)
(C) 450,000 J (D) 112,500 J

Q.22 What is the maximum speed of the roller coaster car as it passes through the valley at ground level ($h = 0$)?

- (A) 20 m/s (B) 30 m/s
(C) 45 m/s (D) 25 m/s

Q.23 What is the speed of the car when it reaches the peak of the second hill (Height B = 25 m)?

- (A) 20 m/s (B) 15 m/s
(C) 25 m/s (D) 10 m/s

Q.24 What is the kinetic energy of the car at the peak of the second hill (Height B)?

- (A) 100 kJ (B) 125 kJ
(C) 75 kJ (D) 225 kJ

Q.25 If realistic track friction dissipates 25 kJ of mechanical energy as heat along the track between Hill A and Hill B, what will be the final kinetic energy at Hill B?

- (A) 100 kJ (B) 75 kJ
(C) 125 kJ (D) 50 kJ

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET C)

01	(A) (B) (C) (D)	02	(A) (B) (C) (D)	03	(A) (B) (C) (D)	04	(A) (B) (C) (D)	05	(A) (B) (C) (D)
06	(A) (B) (C) (D)	07	(A) (B) (C) (D)	08	(A) (B) (C) (D)	09	(A) (B) (C) (D)	10	(A) (B) (C) (D)
11	(A) (B) (C) (D)	12	(A) (B) (C) (D)	13	(A) (B) (C) (D)	14	(A) (B) (C) (D)	15	(A) (B) (C) (D)
16	(A) (B) (C) (D)	17	(A) (B) (C) (D)	18	(A) (B) (C) (D)	19	(A) (B) (C) (D)	20	(A) (B) (C) (D)
21	(A) (B) (C) (D)	22	(A) (B) (C) (D)	23	(A) (B) (C) (D)	24	(A) (B) (C) (D)	25	(A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & EXEMPLAR EXPLANATIONS (SET C)

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.1	(C)	At the top, minimum speed is $v_{\text{top}} = \sqrt{gR}$. By energy conservation: $\frac{1}{2}mv_{\text{bottom}}^2 = \frac{1}{2}mv_{\text{top}}^2 + mg(2R) \Rightarrow v_{\text{bottom}}^2 = gR + 4gR = 5gR \Rightarrow v_{\text{bottom}} = \sqrt{5gR}$.
Q.2	(B)	$SW = \vec{F} \cdot \vec{s} = (3 \times 5) + (4 \times 2) = 15 + 8 = 23$ Joules.
Q.3	(B)	$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(1000)(20^2) = 500 \times 400 = 200,000$ J $= 200$ kJ.
Q.4	(A)	Hooke's law: $F = kx = 500 \text{ N/m} \times 0.04 \text{ m} = 20$ N.
Q.5	(B)	Total energy $S = P \times t = 2000 \text{ W} \times 600 \text{ s} = 1,200,000$ J. $SW = mgh \Rightarrow 1,200,000 = m(10)(10) \Rightarrow 100m = 1,200,000 \Rightarrow m = 12,000$ kg $S = 12,000$ liters.
Q.6	(C)	Friction is a non-conservative, dissipative force; the work done against friction over a closed path is non-zero and dissipates mechanical energy as heat.
Q.7	(B)	Total loss of potential energy $S = mg(h + d) = 2 \times 10 \times (5 + 0.10) = 20 \times 5.10 = 102$ J. Work done by sand $S = F \times d \Rightarrow F \times 0.10 = 102 \Rightarrow F = 1020$ N.
Q.8	(A)	$MA_{\text{ideal}} = \frac{\text{Radius of wheel}}{\text{Radius of axle}} = \frac{40}{8} = 5$.
Q.9	(B)	$p = mv \Rightarrow v = p/m$. $E_k = \frac{1}{2}mv^2 = \frac{1}{2}m(p/m)^2 = \frac{p^2}{2m}$.
Q.10	(A)	Instantaneous power is the scalar product of force and instantaneous velocity: $SP = Fv$.
Q.11	(B)	$\text{Average power} = \frac{\text{Work}}{\text{Time}} = \frac{\frac{1}{2}mv^2}{t} = \frac{1}{2} \left(\frac{mv}{t} \right) v = \frac{1}{2} Fv$.
Q.12	(A)	Input hydraulic power $S = mgh/t = 10,000 \times 10 \times 100 = 10,000,000$ W $= 10$ MW. Output power $S = 0.85 \times 10 = 8.5$ Megawatts.
Q.13	(A)	$SW_{\text{out}} = 400 \times 2 = 800$ J. $SW_{\text{in}} = 100 \times 10 = 1000$ J. $\eta = \frac{800}{1000} \times 100 = 80\%$.
Q.14	(B)	At that height, $SE_p = 50$ J. $mgh = 50 \Rightarrow 1 \times 10 \times h = 50 \Rightarrow h = 5.0$ m.
Q.15	(C)	In a Class 3 lever (e.g. human forearm holding weight, fishing rod, tweezers), effort is applied between fulcrum and load, making effort arm shorter than load arm ($MA < 1$).
Q.16	(A)	Because gravity is conservative, $SW = mgh$ independent of path curvature or inclination.
Q.17	(A)	Since $p = \sqrt{2mE_k}$, the heavier vehicle ($m_{\text{truck}} > m_{\text{car}}$) carries substantially greater momentum.
Q.18	(A)	Energy conservation dictates that a machine can be a force multiplier ($MA > 1$) or a speed multiplier ($VR < 1$), but never both.
Q.19	(A)	$SW = \int P dV$. When volume expands ($dV > 0$), pressure force and boundary displacement align, performing positive work.
Q.20	(D)	Assertion is completely false: In circular orbit, displacement is perpendicular to gravitational force, so zero work is done ($SW=0$) and no fuel is consumed to maintain orbit. Reason is true.
Q.21	(B)	$SE_p = mgh_A = 500 \times 10 \times 45 = 225,000$ J $= 225$ kJ.
Q.22	(B)	$\frac{1}{2}mv^2 = mgh_A \Rightarrow v = \sqrt{2gh_A} = \sqrt{2 \times 10 \times 45} = \sqrt{900} = 30$ m/s (108 km/h).
Q.23	(A)	$v = \sqrt{2g(h_A - h_B)} = \sqrt{2 \times 10 \times (45 - 25)} = \sqrt{20 \times 20} = 20$ m/s.
Q.24	(A)	$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(500)(20^2) = 250 \times 400 = 100,000$ J $= 100$ kJ.
Q.25	(B)	$E_{\text{Available}} = 100 \text{ kJ} - 25 \text{ kJ (friction loss)} = 75$ kJ.